

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to...

1. Identify an appropriate function type for a data set or context by analyzing **rates** of change. (LO 1.13.A)
2. Describe the **assumption** and **domain/range** restrictions embedded in a function model. (LO 1.13.B)

Vocabulary & Key Concepts

first differences:

The differences between consecutive output values of equally-spaced inputs: $\Delta y_i = y_{i+1} - y_i$

second differences:

Differences of the first differences; constant for quadratic data......

domain restriction:

A limit on the input values required by mathematics, context, or practical meaning......

piecewise-defined function:

A function defined by different formulas over different intervals of its domain......

model assumption:

A simplifying claim about the real world that must hold for the model to be valid......**Hook**

“A scientist hands you a table of mystery data. How do you decide which function type to use?”

Without graphing, you can inspect the **differences** to detect the function’s degree.**Rate-of-Change Clues: The Difference-Table Method****Setup:** Inputs must be **equally**-spaced. Compute differences of the output column.

x	y	Δy (1st diff.)	$\Delta^2 y$ (2nd diff.)
0	1	—	—
1	4	3	—
2	11	7	4
3	22	11	4
4	37	15	4

Decision tree:

- 1st differences **constant** $\Rightarrow$ model is **linear** (degree **1**)
- 2nd differences **constant** $\Rightarrow$ model is **quadratic** (degree **2**)
- 3rd differences **constant** $\Rightarrow$ model is **cubic** (degree **3**)
- n th differences constant $\Rightarrow$ model is degree- n polynomial

Key idea: Constant *first* differences mean a **constant** rate of change. Constant *second* differences mean the rate of change is itself **linear (changing at a constant rate)**.

Teacher Note

The example table above has constant 2nd differences of 4, so the data is quadratic. Students should notice 1st differences are not constant, prompting them to compute 2nd differences.

Shape and Context Clues

When you have a description but no table, use context:

Context clue	Best model type
Roughly constant rate of change	linear
Area or 2-D measurement	quadratic
Volume or 3-D measurement	cubic
Multiple local maxima/minima	higher-degree polynomial
Behavior changes at a threshold	piecewise-defined

Special rule: A degree- n polynomial can model a data set with exactly $n + 1$ points (with distinct inputs).

Piecewise Models

A piecewise-defined function uses **different** formulas over **different** intervals.

Example scenario: A delivery driver travels at a constant speed for the first 20 minutes, stops for 5 minutes, then accelerates home. This situation requires **3** pieces because the **behavior / rate of change** changes over different intervals.

To describe a piecewise model, state:

1. The **formula (or function type)** for each piece.
2. The type of function for each piece (e.g., **linear**, **constant**).
3. The **domain / interval** conditions (where each piece applies).

Stating Assumptions and Restrictions

Every model embeds assumptions. Fill in the sentence frame:

“This model assumes that **the relationship between variables** remains **consistent / proportional** over **the modeled interval**. The domain is restricted to **values where the context is meaningful** because **negative or extreme inputs have no real-world interpretation**.”

Example: A quadratic models the area A of a rectangle with a fixed perimeter of 120 ft, where ℓ is one side length.

Assumption: **The shape is a perfect rectangle and the perimeter is exactly 120 ft.**

Domain restriction: **$0 < \ell < 60$ (length must be positive and less than half the perimeter).**

Range restriction: **$0 < A \leq 900$ (area is positive; maximum at $\ell = 30$).**